Title of Project:
Fetal brain responses to light and sound measured with optically pumped magnetometers

You are invited to take part in a research study. Before you decide, it is important for you to understand why the research is being done and what it involves. Please take time to read the following information carefully and discuss it with others, if you wish. Ask us if there is anything that is not clear or if you would like more information. We can present information about the study in person or by video call, if desired. Take time to decide whether or not you wish to take part. Thank you for reading this and considering our study.

What is the purpose of this study?
This study aims to measure the heart and brain activity of the developing human fetus, using a new magnetic sensor technology. The main goal will be to measure the fetal brain responses to sound and light.
We hope these measurements will serve as a proof-of-concept for using these sensors in future experiments to study how the human brain develops in greater detail. Once the technique is established, it may also prove useful for monitoring fetal brain development in high-risk pregnancies.

Are there any benefits to participating in the study?
You will receive a fetal ultrasound scan and, if desired, a print of the scan. Participation in the study will not affect your normal prenatal or medical care.

You will receive 150 DKK/hour for participation in the study. You will therefore be compensated 300 DKK for each typical session of 2 hours. This compensation is taxable as B-income for residents of Denmark.

Who can take part?
You can take part in this study if:
• you are pregnant and 18 years of age or older
• you are in good health
• your pregnancy has not had complications or special risk factors
• you are available during the specific weeks of pregnancy relevant to the study (all possibilities after week 25)

What will happen to me and what is expected if I take part?
The experiment consists of either a single session, or 3 sessions approximately 4 weeks apart, with a maximum duration of 2 hours per session. Before we begin, we will summarize how the experiment will go, and you can ask any remaining questions you might have. You are welcome to bring a person of your choice to accompany you to the laboratory session.

You will be asked to change into hospital clothing after arrival. You will be asked to remove items that may contain metal, for example an underwire bra and any jewelry (rings, earrings, piercings, etc.), so you may prefer to leave these at home.

First, you will receive an ultrasound scan to determine the position of the fetus within the womb and estimate the fetal heart rate. This scan will be very similar to the routine ultrasound scan you will have already done as part of your prenatal care.

After this, you will be helped into the MEG (magnetoencephalography) cabin. You will be asked to lay down on the bed or sit down in the chair inside the cabin. If the experiment will use light flashes, then you will be asked to expose your abdomen for the duration of the recording, so that the light can reach your skin and your baby inside.

Optically pumped magnetometers, small magnetic sensors, will be placed on your abdomen to measure your baby's heart and brain activity. The sensors can also capture your own heart activity. The sensors should feel pleasantly warm. We will ask you to refrain from movements as much as possible during the recording, so we will work with you to ensure the most comfortable position. We can provide extra pillows or a blanket, for example.

We may take photos to document the position of the sensors around your body.

The lights will then be dimmed and we will close the door. The cabin is sound insulated, but we will be able to hear you through a microphone and communicate with you through the intercom. For this experiment, you will be asked to relax and remain in the same position to the extent possible – you may sleep if you like!

We will attempt to measure your baby's brain activity in response to light flashes and/or brief sounds. The light flashes and sounds do not exceed the intensity of everyday exposure. You may be provided with earphones playing low-level white noise or your desired playlist. We will also make measurements during rest with no external stimulation.

The MEG recording will take a maximum of 100 minutes, but typically 70-80 minutes in total.

What are the devices involved?
We will place *optically pumped magnetometers*, which are ultra-sensitive magnetic sensors, on your abdomen to measure your baby's heart and brain activity, as well as your own heart activity.

What are the possible risks or side effects of taking part?
There are no safety risks posed by the experiment or the sensors. The sensors operate at low voltages, are completely sealed in plastic, and do not use radiation, chemicals, or strong magnets. Some participants may find the MEG cabin claustrophobic, but please remember that you can communicate with us and ask to take a break or stop completely if you become uncomfortable.

The magnetic measurements are purely for scientific purposes and are not optimized to detect clinical abnormalities. Ordinarily the data will not be reviewed by qualified medical personnel. Nevertheless, there is a remote possibility that the magnetic measurements or ultrasound may suggest a previously unknown medical concern. In this case, we will consult with an appropriate clinical staff member, who will inform you if further medical consultation is recommended.

Confidentiality - who will have access to the data?
Your participation is confidential and will not be included in your medical record. No personally identifying information (such as your name or CPR number) is ever stored with the data, though basic demographic information such as your age and week of pregnancy will be. Fully anonymized data may be made available in public databases, with your consent. Any photographs will not be included in such public databases. Processed results may be published or presented as discussed in the next section.

What happens at the end of the study?
Fully anonymized data and the study results may be published in a journal, included in grant applications, or used for teaching purposes. With your consent, they may also be included in public databases. The scientific results may also be presented at scientific meetings or in talks at academic institutions. Results will always be presented in a way that ensures that individual participants cannot be identified.

Can I ask questions about the research project?
You may ask more questions about the study at any time - before, during and after the study. The contact details of the investigators are provided above, so that they are available to answer your questions or concerns about the study. Please do not hesitate to ask the experimenter if there are any points about the study that remain unclear to you. You are also welcome to invite a person of your choice to join for meetings or discussions about the project.

Can I withdraw from the study?
Your participation in this research project is voluntary, and you may withdraw from the research at any time and for any reason, without explaining why, and this will not affect your legal rights.

Can the investigators interrupt the study?
The study may likewise be interrupted by the investigators at any time, for any variety of reasons, for example technical difficulties or a change in the eligibility criteria.

Who has funded the study?

The Lundbeck Foundation has funded the project with a grant of 5.1M DKK for personnel, laboratory costs, and participant compensation [to be revised to 8.1M DKK if the additional postdoc fellowship is funded]. The Carlsberg Foundation has funded the MEG sensors used for this project with a grant of 3M DKK. The researchers do not have any financial ties to either foundation. The funders have no influence on the study's design, analysis, or reporting; all scientific decisions are made independently by the research team.

What other rights do I have?
If you are participating in a health science research project, it is important that you are aware of your rights. You can read about them below.

As a participant in a health science research project, you should know that:

1. Your participation in the research project is entirely voluntary and can only occur after you have received both written and oral information about the research project and signed the consent form.
2. You can withdraw your consent to participate at any time, verbally, in writing, or by any other clear indication, and exit the research project. If you withdraw your consent, it does not affect your right to current or future treatment or other rights you may have.
3. You have the right to bring a family member, friend, or acquaintance to the information meeting.
4. You have the right to a reflection period before signing the consent form.
5. Information about your health conditions, other purely private matters, and other confidential information about you that emerges in connection with the research project is subject to confidentiality.

6. The processing of information about you, including information in your blood samples and tissue, is done according to the rules in the General Data Protection Regulation, the Data Protection Act, and the Health Act. The data controller in the trial must inform you more about your rights under the data protection rules.
7. There is the possibility to access trial protocols under the provisions of the Public Access to Information Act. This means you can access all papers regarding the arrangement of the trial, except for parts containing trade secrets or confidential information about others.
8. There is the possibility to complain and receive compensation according to the rules in the law on complaints and compensation access within the healthcare system. If an injury occurs during the trial, you can contact the Patient Compensation Association, see more at www.patienterstatningen.dk

How can I contact the project leaders?
If you have any questions, you are very welcome to contact us:

Professor Sarang Dalal
Center of Functionally Integrative Neuroscience
Aarhus University
Universitetsbyen 3, Building 1710
8000 Aarhus C
email: sarang@cfin.au.dk
phone: +45 9350 8443

Professor and Consultant Physician Lars Henning Pedersen
Aarhus University Hospital
Gynecology and Obstetrics
Palle Juul-Jensens Boulevard 99
8200 Aarhus N
email: lars.h.pedersen@auh.rm.dk
phone: +45 5052 6512